Calculus I

Tutorial #1 (30 Sep - 4 Oct 2024)

Topics:

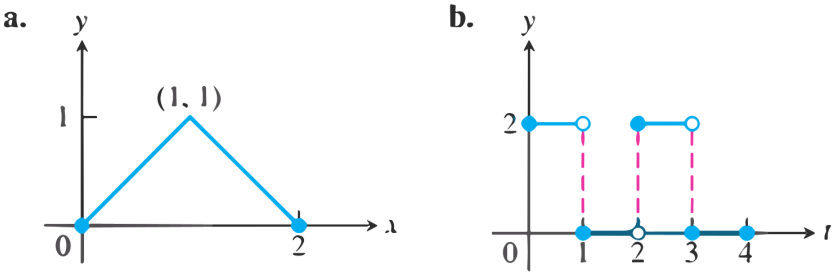
- Functions and their graph
- Combining functions
- Trigonometrical functions

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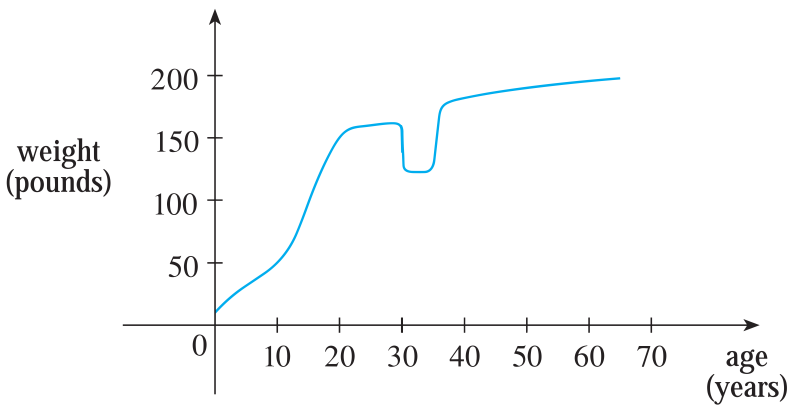
Problem 1. Graph the following equations and explain why they are not graphs of functions of x.

a. $|y| = x$ **b.** $y^2 = x^2$

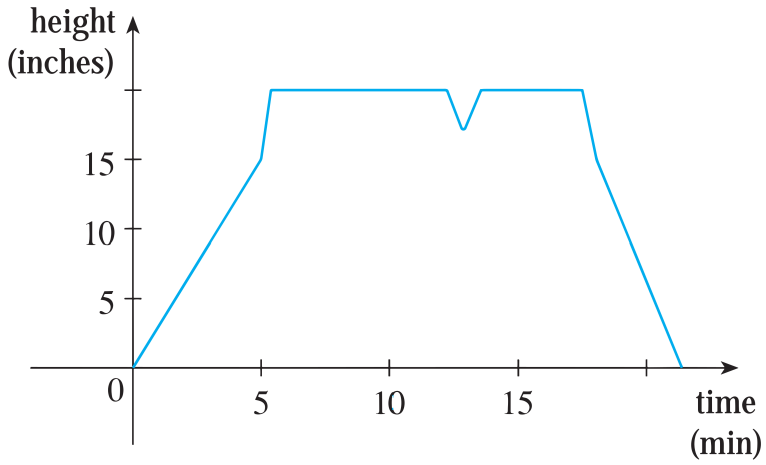
Problem 2. Find a formula for each function graphed



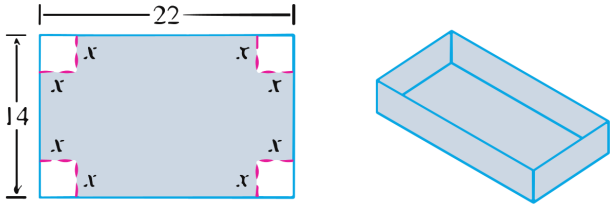
Problem 3. The graph shown gives the weight of a certain person as a function of age. Describe in words how this person's weight varies over time. What do you think happened when this person was 30 years old?



Problem 4. The graph shows the height of the water in a bathtub as a function of time. Give a verbal description of what you think happened.

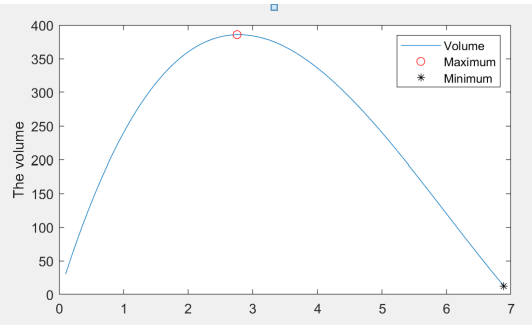


Problem 5. A box with an open top is to be constructed from a rectangular piece of cardboard with dimensions 14 in. by 22 in. by cutting out equal squares of side x at each corner and then folding up the sides as in the figure.

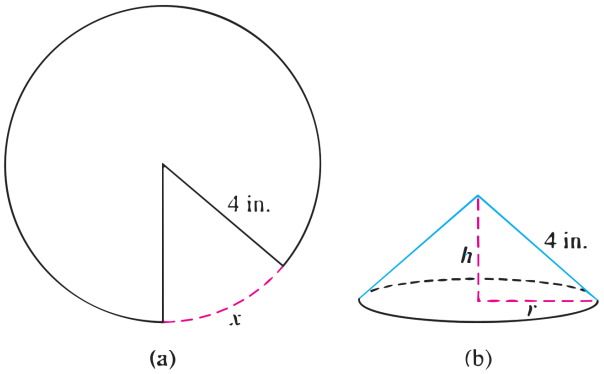


a) Express the volume V of the box as a function of x and find the domain of the function.

b) Observing the graph of the volume function obtained from the previous step: Estimate the range of V, describle in words what the graph tells you about the volume.



Problem 6. *(A cone problem)* Begin with a circular piece of paper with a 4 in. radius as shown in part (a). Cut out a sector with an arc length of x. Join the two edges of the remaining portion to form a cone with radius r and height h, as shown in part (b).



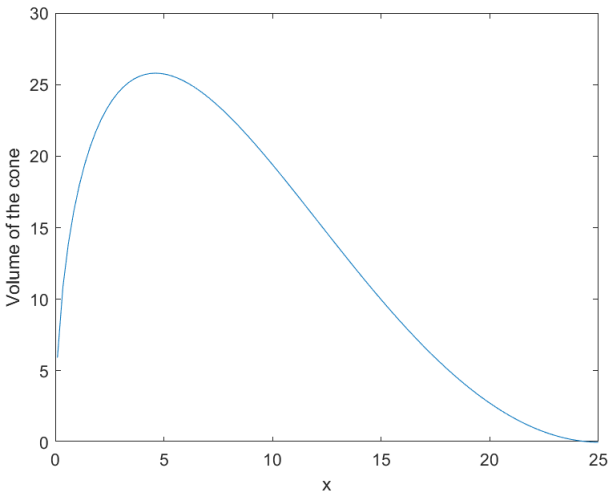
a) Explain why the circumference of the base of the cone is $8\pi - x$.

b) Express the radius r as a function of x.

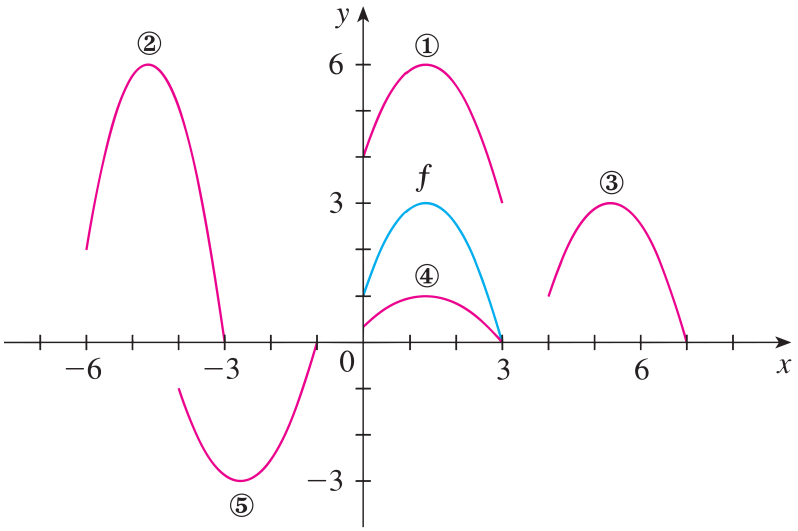
c) Express the height h as a function of x.

d) Express the volume V of the cone as a function of x.

e) Given the following graph of $V(x)$, can we say something about the behavior of the volume?



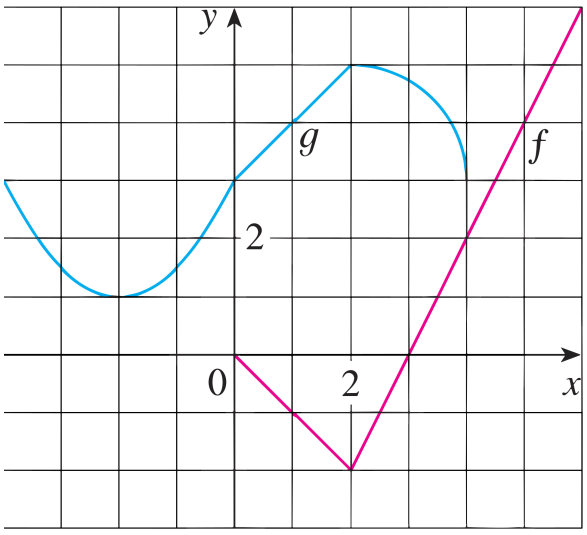
Problem 7. The graph of $y = f(x)$ is given. Match each question with its graph and give reasons for your choices.
a) $y = f(x-4)$. **b)** $y = f(x)+3$. **c)** $y = \frac{1}{3}f(x)$. **d)** $y = -f(x+4)$.
e) $y = 2f(x+6)$.



Problem 8. Use the given graphs of f and g to evaluate each expression, or explain why it is undefined.

a) $f(g(2))$; **b)** $g(f(0))$; **c)** $(f \circ g)(0)$.

d) $(g \circ f)(6)$; **e)** $(g \circ g)(-2)$; **f)** $(f \circ f)(4)$.



Problem 9. If you invest x dollars at 4% interest compounded annually, then the amount $A(x)$ of the investment after one year is $A(x) = 1.04x$. Find $A \circ A$, $A \circ A \circ A$ and $A \circ A \circ A \circ A$. What do these compositions represent? Find a formula for the composition of n copies of A.

Problem 10. Graph the equation $|x| + |y| = 1 + x$.

THE END